Harish Manoharan

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Software engineer at Purgo AI working on agents, retrieval and LLM evaluation. M.S. Computer Science, UT Dallas, 2026.

EXPERIENCE

Purgo AI San Francisco Bay Area
AI startup: an agent that turns data-engineering tickets into PySpark, SQL and dbt code
Software Engineer Jun 2026 – Present
Software Engineer Intern (on site, then remote during M.S.) Jun 2025 – May 2026

- Cut missed source tables from 79 to 24 on our 103-ticket internal benchmark by adding dbt lineage expansion (SQL parsing with sqlglot) to the LangGraph agent's retrieval, with no ticket losing a table it had already found.
- Built Purgo's IQ/OQ validation engine (NestJS/TypeScript, GCP Cloud Run), which qualifies life-sciences (GxP) customers' Databricks platforms on Azure and AWS: it runs a git-versioned library of 59 tests as cancellable jobs and keeps redacted evidence for signed audit reports; an LLM pipeline turns requirement documents into new tests.
- Built the LLM validation product across UI, API and test engine (React/Next.js, NestJS, PostgreSQL): 10 behavioral tests on Azure OpenAI or Databricks-served models (hallucination on missing fields, extraction F1, demographic bias, 20-run determinism, latency at 20 concurrent calls) and 9 checks on an Azure deployment's access, network and audit setup.
- Turned recurring Databricks runtime failures into AST lint rules that the agent's self-review runs on generated PySpark/SQL, so those errors surface in review and runtime failures dropped; added dbt source checks and a recovery pass that keeps the candidate passing the most checks when all candidates fail.
- Built Pinecone semantic search over Polaris/Iceberg catalogs (re-indexes only tables whose schema changed, keyword fallback) and its 188-query benchmark: recall@10 0.68 vs 0.54, mean latency 0.18 s vs 4.0 s for the existing search; built the table sampling that masks PII/PHI with Presidio before rows reach the model.
- Rewrote attachment chunking to keep records whole and tell the model what was left out; moved two agent stages to a smaller model after five 102-ticket benchmark runs showed no loss in retrieval recall (those stages 7.6x cheaper).

Digitus Business Solutions Bengaluru, India
Software Engineer Intern May 2024 – Jul 2024

- Raised SQL execution accuracy from 52% to 76% on a 500-query evaluation set by building the text-to-SQL agent behind a database chatbot used mainly by the sales team, with schema discovery and retries driven by execution errors.
- Built the API integration between QAD ERP and Malaysia's MyInvois e-invoicing system, which clients use to submit invoices to the tax authority and get notified of document events.

PwC India Bengaluru, India
Technology Consulting Intern, Data Analytics May 2023 – Jul 2023

- Built a customer analytics platform served through Django REST APIs and Power BI; its XGBoost + SARIMA demand forecasts cut MAPE 15% against the baseline across 50+ SKUs.

EDUCATION

The University of Texas at Dallas M.S. Computer Science May 2026
GPA 3.9/4.0 · Dean's Graduate Scholar

Indian Institute of Technology Madras Dual Degree: M.Tech Data Science, B.S. Biological Sciences May 2024
INSPIRE Scholar · Young Research Fellow · Thesis with The Jackson Laboratory on unsupervised behavior discovery in mice

SKILLS

Languages: Python, TypeScript, JavaScript, SQL
Machine learning and LLMs: retrieval-augmented generation (RAG), LLM evaluation, AI agents (LangGraph), PyTorch
Data: Databricks, Apache Spark (PySpark), dbt, PostgreSQL, Apache Iceberg
Web, backend and cloud: React, Next.js, FastAPI, NestJS, REST APIs, AWS, Azure, GCP, Docker, CI/CD (GitHub Actions)